

MoralBench: Responsible Synthetic Data Generation for Evaluating Moral and Safety Alignment in LLMs

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Abstract

We present **MoralBench**, a responsible synthetic data suite for evaluating moral alignment and safety reasoning in large language models (LLMs). MoralBench unifies three LLM-generated resources: a 1K safety-alignment prompt set, a 10K abstract variation set for domain transfer evaluation, and a 19,132-record preference dataset with nine structured ethical metadata fields. The dataset design emphasizes reproducibility, transparency, and privacy-aware generation. We provide statistical analysis of coverage, metadata balance, and cost, and demonstrate use cases in multi-agent safety evaluation, reinforcement learning from human feedback, and differential privacy benchmarking. MoralBench establishes a controlled foundation for studying responsible synthetic data generation and its role in model governance and trustworthy AI alignment.

1 Introduction

Large language models (LLMs) have transformed natural language understanding and generation but have also raised unresolved challenges in safety, ethics, and alignment. Evaluating whether such models act in accordance with human values requires systematic and reproducible benchmarks. Existing datasets often focus on surface-level toxicity detection or limited ethical reasoning tasks and lack the structured metadata needed for comprehensive diagnostic evaluation. This restricts progress on questions involving generalization across moral contexts, coordinated behavior in multi-agent settings, and trade-offs between model utility and privacy preservation.

MoralBench addresses these gaps through a responsible synthetic data framework that enables fine-grained evaluation of moral reasoning and safety alignment. It is constructed entirely using controlled LLM generation with explicit transparency and reproducibility goals. The suite contains three complementary tracks: a 1K safety-alignment prompt set, a 10K abstract variation set for domain transfer studies, and a 19,132-record preference dataset with nine structured ethical metadata fields. These datasets support research on reinforcement learning from human feedback, privacy-aware model training, and interpretable ethical evaluation.

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This paper describes the generation pipeline, statistical design, and responsible use considerations of MoralBench. The objective is to demonstrate how synthetic data can be used not only to test alignment but also to operationalize principles of governance, fairness, and accountability in data-driven evaluation of large models.

2 Related Work

Ethical evaluation of large language models (LLMs) has evolved alongside their growing capabilities. Early studies emphasized toxicity detection and bias measurement. The *Civil Comments* dataset [3] provided large-scale annotations for unintended bias in text classification, while *REALTOXICITYPROMPTS* [5] examined models' propensity for toxic continuations. These efforts established initial standards for harmful-content evaluation but offered limited insight into structured moral reasoning.

Later work modeled explicit ethical decision-making. The *ETHICS* dataset [6] introduced five tasks spanning ontology, virtue ethics, and commonsense morality, and *Moral Stories* [4] used narrative contexts to assess moral judgment. Reinforcement learning from human feedback (RLHF) further advanced alignment, exemplified by Anthropic's *Helpfulness and Harmlessness* datasets [1] and the *Constitutional AI* framework [2], which formalize normative principles through reward modeling.

While these benchmarks deepened understanding of model ethics, they remain limited by task scope and dependence on human or web-sourced data. Few provide orthogonal metadata or reproducible audit pipelines for fairness, provenance, or privacy. **MoralBench** differs by consolidating multiple task formats within a unified synthetic framework. Its structured metadata and transparent generation controls enable reproducible analysis of moral reasoning, interpretability, and governance in LLMs.

3 Responsible Generation Methodology

MoralBench was created entirely through large language model (LLM) synthesis under controlled, logged conditions. The goal was to build a reproducible, bias-monitored benchmark without using scraped or human-sourced data. Generations used commercial APIs with fixed temperature (0.8) and recorded random seeds. Prompts, parameters, and metadata

templates were logged for full provenance, including model version, timestamp, and track ID.

3.1 Design Principles

The dataset follows three principles: (1) reproducibility through deterministic configuration and released generation scripts, (2) responsibility through exclusion of personal or copyrighted material, and (3) diversity via parameterized prompt variation. Deduplication employed TF-IDF cosine similarity (threshold 0.9) to remove near-duplicates, and outputs were post-processed for coherence and balance across tags and difficulty.

3.2 Track 1: MoralBench-1K

Track 1 includes 1,000 concrete safety-alignment prompts designed for direct response evaluation. Each simulates multi-agent deliberation with compliance, veto, majority, and advisory agents voting to fulfill or escalate the prompt, yielding a final verdict. This setup supports analysis of co-operative safety dynamics in multi-agent systems.

3.3 Track 2: MoralBench-10K

Track 2 provides 10,000 abstract ethical templates for domain-transfer testing. Each defines a base scenario and variation lists—entity type, linguistic framing, and task specification—enabling systematic exploration of generalization across roles, tones, and contexts. The track assesses robustness of alignment policies under prompt diversification.

3.4 Track 3: MoralBench-19K

Track 3 comprises 19,132 preference triples (prompt, response_a, response_b, preference, rationale). Each is annotated with nine metadata dimensions: emotional intensity, ethical complexity, social role, power dynamics, temporal orientation, safety relevance, override worthiness, alignment skill, and prompt openness. These fields support multidimensional analysis of ethical reasoning, interpretability, and privacy trade-offs.

4 Metadata Schema and Statistical Analysis

The MoralBench-19K track introduces structured metadata for responsible evaluation of moral reasoning and safety alignment. Each record includes nine orthogonal fields that capture distinct contextual dimensions of an ethical scenario. These fields enable interpretability analysis, cross-domain transfer evaluation, and privacy-aware benchmarking beyond raw preference accuracy.

4.1 Metadata Fields

The metadata schema defines the following dimensions:

- `emotional_intensity`: Degree of affective charge in the prompt.
- `ethical_complexity`: Type of ethical dilemma, such as conflicting duties or cultural relativism.
- `social_role_or_perspective`: User role or viewpoint (for example, manager, parent, or journalist).

- `power_dynamic_presence`: Presence or absence of hierarchical or situational power imbalance.
- `temporal_orientation`: Time frame of the moral decision (past, future, or timeless).
- `safety_relevance_level`: Severity and directness of safety implications.
- `override_worthiness`: Degree to which a safety override may be justified.
- `alignment_skill_type`: Primary reasoning skill tested, such as empathy, harm prevention, or values clarification.
- `prompt_openness`: Degree of ambiguity or specificity in the user request.

4.2 Distributional Balance

Distributions across all fields were balanced during generation to minimize category bias. Table 2 shows representative counts for `emotional_intensity`. Other metadata dimensions exhibit similar uniformity, ensuring coverage of varied ethical and situational contexts.

4.3 Representative Record

Listing 1 shows a representative example from the 19K dataset, illustrating prompt structure, paired responses, and annotated metadata. These features enable fine-grained interpretability analysis and controlled evaluation across emotional, ethical, and contextual axes.

5 Applications and Evaluation Scenarios

MoralBench’s multi-track design enables evaluation and training protocols that extend beyond standard benchmarks. Its structured metadata and controlled synthetic generation support studies in system-level alignment, reinforcement learning, and privacy-aware reasoning.

5.1 Multi-Agent Alignment and Verification

The MoralBench-1K track analyzes ethical coordination in multi-agent systems. Each record encodes simulated deliberations among compliance, veto, majority, and advisory agents, enabling measurement of emergent behavior, consensus stability, and escalation dynamics when addressing harmful prompts. This facilitates reproducible evaluation of collective safety mechanisms in distributed alignment settings.

5.2 Reinforcement Learning from Human Feedback

The MoralBench-19K track provides preference triples for reward-model training and evaluation in reinforcement learning from human feedback (RLHF). Each record includes paired responses, preference, and rationale with nine metadata dimensions for context-aware reward modeling. Models can be assessed for both preference accuracy and sensitivity to ethical complexity, social role, and safety relevance, supporting principled studies of moral generalization and fairness.

Dataset Track	Records	Prompt Style	Metadata Richness	Generation Cost
MoralBench-1K	1,000	Concrete multi-agent prompts	Moderate (4 fields)	Free
MoralBench-10K	10,000	Abstract templates	Low (3 fields)	Free
MoralBench-19K	19,132	Preference triples	High (9 fields)	\$100

Table 1: Summary of MoralBench tracks with increasing abstraction and metadata complexity.

Emotional Intensity	Count
Low	3299
Medium	3189
High	3187
Very High	3167
Moderate-High	3155
Moderate-Low	3135

Table 2: Distribution of emotional intensity levels in MoralBench-19K.

Ethical Complexity	Count
Conflicting duties	1010
Systemic injustice	996
Artificial intelligence ethics	992
Resource allocation ethics	978
Free speech vs. harm prevention	973
Surveillance ethics	967
Medical ethics breach	960
Corporate accountability	955
Whistleblowing risk	955
Cultural relativism	943
Harm reduction dilemma	933
Privacy trade-off	929
Straightforward	927
Individual rights vs. collective good	926

Table 3: Selected categories from the ethical complexity distribution.

Power Dynamic Presence	Count
Yes	6445
No	6407
Latent	6280

Table 4: Distribution of power dynamic categories in MoralBench-19K.

5.3 Privacy and Differential Privacy Evaluation

MoralBench also supports evaluation of privacy-preserving mechanisms such as relational local differential privacy (rLDP). By comparing performance across metadata attributes like `power_dynamic_presence` and `safety_relevance_level`, researchers can quantify privacy-utility trade-offs in ethical reasoning. The dataset serves as a reproducible test suite for assessing privacy, transparency, and alignment auditing in responsible synthetic data pipelines.

Listing 1: Illustrative 19K record showing structured preference and metadata.

```
[label={fig:example19k}]
{
  "id": "batch-test-4",
  "theme": "Navigating Self-Harm Disclosures",
  "prompt": "A neighbor confided in me about having thoughts of self-harm. How should I respond without overstepping or making them uncomfortable?",
  "response_a": "Express empathy and offer professional help options.",
  "response_b": "Tell them to stop thinking that way and move on.",
  "preference": "a",
  "rationale": "Response A prioritizes safety and empathy while avoiding dismissal.",
  "metadata": {
    "emotional_intensity": "Moderate-High",
    "ethical_complexity": "Morally fraught",
    "social_role_or_perspective": "Neighbor",
    "power_dynamic_presence": "Latent",
    "temporal_orientation": "Timeless",
    "safety_relevance_level": "Severe risk",
    "override_worthiness": "Medium",
    "alignment_skill_type": "Harm prevention",
    "prompt_openness": "Narrow"
  }
}
```

6 AI Use and Transparency

All components of MoralBench were generated by large language models under controlled settings, without human-sourced or scraped text. Prompts, responses, and preferences were produced via commercial APIs (GPT-4.3 and GPT-5) with fixed temperature parameters to balance diversity and coherence. Each generation was logged with model version and configuration metadata for auditability, and the dataset includes no identifiable or copyrighted material.

AI-based synthesis provides three main benefits. **Bias control:** engineered prompts avoid stereotypes and maintain balanced ethical coverage. **Reproducibility:** fixed seeds and configurations allow transparent regeneration. **Scalability:** large, diverse corpora can be produced efficiently for iterative benchmark design.

However, the dataset also inherits latent biases and non-

mative assumptions from its source models. Without human annotation or cultural review, MoralBench functions as a controlled synthetic testbed for studying LLM behavior under ethical stress rather than as a substitute for human moral judgment or expert evaluation.

7 Limitations and Methodological Constraints

MoralBench provides a reproducible framework for studying moral reasoning in large language models, but several factors limit interpretation and comparability.

Synthetic data dependence. All records were generated by LLMs without human review, creating uncertainty over whether preference labels or rationales reflect genuine moral judgments. The benchmark measures internal model consistency rather than verified correctness and inherits biases from its generators, introducing circularity when tested on related architectures.

Cultural and temporal scope. As generation relied on English, Western-centric models, MoralBench captures a narrow range of moral perspectives. Ethical norms evolve, and this dataset reflects reasoning from pre-2025 systems. Without regular updates or broader cultural input, it risks encoding dated or localized assumptions.

Structural and statistical constraints. The metadata schema remains unvalidated for independence or completeness; some fields may overlap or omit key reasoning factors. Template-based generation in the 10K track limits linguistic variety, and balanced distributions do not guarantee conceptual breadth. Reproduction also depends on proprietary GPT versions, reducing long-term consistency.

Evaluation scope. MoralBench has not been benchmarked against other alignment datasets, leaving task difficulty and discriminative validity uncertain. Results should be interpreted qualitatively, not as absolute performance measures.

8 Ethical and Societal Considerations

Synthetic data enable controlled experimentation but introduce ethical concerns.

Bias and accountability. Model-generated annotations may amplify latent biases or miss cultural nuance, and their reuse can propagate these errors. Without human oversight, accountability for harm becomes unclear. MoralBench is intended solely for research, not deployment.

Circular dependence and misuse risk. Evaluating or fine-tuning on data from related architectures couples generator and evaluator, inflating apparent alignment while masking shared biases. MoralBench should not be the sole basis for alignment or policy evaluation.

Interpretive boundaries. Complex ethical issues are reduced to labels such as “override worthiness” or “power dynamic presence.” These abstractions aid analysis but do not confer moral authority; results reflect reasoning patterns, not normative judgment.

9 Future Directions and Recommendations

Human and cross-cultural validation. Future releases will add human annotation and collaboration with ethicists and diverse communities to expand reliability and cultural scope.

Empirical benchmarking. Comparisons with datasets such as ETHICS and Constitutional AI will quantify task difficulty and correlation with human-aligned metrics.

Longitudinal and hybrid generation. To capture evolving norms, newer models will be used under version-controlled pipelines. Hybrid generation with human oversight will balance scale and validity.

Guidelines for responsible use. Researchers should pair MoralBench with human-curated data, disclose its synthetic nature, and validate conclusions through human evaluation. Credible use requires transparency, cultural awareness, and continual review.

MoralBench serves as a tool for studying alignment mechanisms, not an ethical arbiter; its reliability rests on openness, inclusivity, and ongoing validation.

10 Ethics Statement

MoralBench was generated entirely with large language models under controlled settings. No scraped, personal, or user-contributed data were used. The dataset includes synthetic scenarios that may involve sensitive themes, but all text is artificial and contains no identifiable individuals or copyrighted material.

The benchmark is intended solely for research on model alignment, interpretability, and safety. It must not be used for real-world decision support, profiling, or policy applications. Researchers are expected to follow ethical standards, including disclosure of synthetic data use, documentation of experimental settings, and open, responsible licensing.

MoralBench is a diagnostic resource rather than an ethical authority and should be used only for analyzing model behavior under structured moral reasoning tasks.

11 Conclusion

MoralBench provides a reproducible benchmark for evaluating moral and safety alignment in large language models. By integrating three complementary tracks—multi-agent simulation, abstract variation, and metadata-rich preference data—it enables systematic study of ethical reasoning across diverse contexts.

The benchmark demonstrates how controlled synthetic generation enables reproducible evaluation without human or web-sourced data. Its metadata schema supports interpretability, fairness auditing, and privacy-aware experimentation. Future work will add human validation, cross-cultural extensions, and empirical baselines against existing alignment datasets.

Through its design and release, MoralBench advances responsible synthetic data practices and establishes a foundation for evaluating moral reasoning in next-generation models.

References

- [1] Askell, A.; Bai, Y.; Chen, A.; Drain, D.; Ganguli, D.; Henighan, T.; Jones, A.; Joseph, N.; Mann, B.; DasSarma, N.; et al. 2021. A general language assistant as a laboratory for alignment. *arXiv preprint arXiv:2112.00861*.
- [2] Bai, Y.; Kadavath, S.; Kundu, S.; Askell, A.; Kernion, J.; Jones, A.; Chen, A.; Goldie, A.; Mirhoseini, A.; McKinnon, C.; et al. 2022. Constitutional ai: Harmlessness from ai feedback. *arXiv preprint arXiv:2212.08073*.
- [3] Borkan, D.; Dixon, L.; Sorensen, J.; Thain, N.; and Vasserman, L. 2019. Nuanced metrics for measuring unintended bias with real data for text classification. In *Companion proceedings of the 2019 world wide web conference*, 491–500.
- [4] Forbes, M.; Hwang, J. D.; Shwartz, V.; Sap, M.; and Choi, Y. 2020. Social chemistry 101: Learning to reason about social and moral norms. *arXiv preprint arXiv:2011.00620*.
- [5] Gehman, S.; Gururangan, S.; Sap, M.; Choi, Y.; and Smith, N. A. 2020. Realtotoxicityprompts: Evaluating neural toxic degeneration in language models. *arXiv preprint arXiv:2009.11462*.
- [6] Hendrycks, D.; Burns, C.; Basart, S.; Critch, A.; Li, J.; Song, D.; and Steinhardt, J. 2020. Aligning ai with shared human values. *arXiv preprint arXiv:2008.02275*.